ELECTRICAL MACHINES · PAPER

Jun 2023 exam: full worked solutions

Answers and worked solutions · Day 18 · Tue 10 Nov · pack MX1

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Battery Modelling [5 marks]

(i) With no current there is no drop across R_S, so

$$E = V_{oc} = \mathbf{13.8\ V}$$

(ii) KVL around the loop: $V = E - IR_S$, so

$$R_S = \frac{E - V}{I} = \frac{13.8 - 9.5}{300} = \mathbf{0.01433\ \Omega}\ (14.33\ \text{m}\Omega)$$

$$P_{R_S} = I^2 R_S = 300^2 \times 0.01433 = \mathbf{1290\ W}$$

$$P_{motor} = VI = 9.5 \times 300 = \mathbf{2850\ W}$$

Check: $EI = 13.8 \times 300 = 4140$ W $= 1290 + 2850$ W.

(iii)

$$V = E - IR_S = 13.8 - 20 \times 0.01433 = \mathbf{13.51\ V}$$

1b) Star-Delta Equivalent Circuits [6 marks]

(i) Star. Each resistor sees the phase voltage.

$$V_1 = \frac{V_L}{\sqrt{3}} = \frac{460}{\sqrt{3}} = 265.6\ \text{V}$$

$$I_1 = \frac{V_1}{R} = \frac{265.6}{70} = \mathbf{3.794\ A}, \qquad I_L = I_1 = \mathbf{3.794\ A}$$

$$P = 3V_1 I_1 = 3 \times 265.6 \times 3.794 = \mathbf{3.023\ kW}$$

(ii) Delta. Each resistor sees the full line voltage.

$$I_1 = \frac{V_L}{R} = \frac{460}{70} = \mathbf{6.571\ A}, \qquad I_L = \sqrt{3}\, I_1 = \mathbf{11.38\ A}$$

$$P = 3V_L I_1 = 3 \times 460 \times 6.571 = \mathbf{9.069\ kW}$$

Check: the delta power is exactly 3 times the star power ($9.069/3.023 = 3$), as it must be for the same resistors.

1c) Phasor Calculations [9 marks]

Multiply magnitudes and add angles. $j = 1\angle 90^\circ$ and $-j = 1\angle -90^\circ$.

(i)

$$\mathbf{V}_R = R\,\mathbf{I}_1 = 10 \times 13\angle 20^\circ = \mathbf{130\angle 20^\circ\ V}$$

$$\mathbf{V}_L = jX_L\,\mathbf{I}_1 = 30\angle 90^\circ \times 13\angle 20^\circ = \mathbf{390\angle 110^\circ\ V}$$

$$\mathbf{V}_C = -jX_C\,\mathbf{I}_1 = 50\angle -90^\circ \times 13\angle 20^\circ = \mathbf{650\angle -70^\circ\ V}$$

(ii)

$$Z = R + jX_L - jX_C = 10 + j30 - j50 = 10 - j20 = \mathbf{22.36\angle -63.43^\circ\ \Omega}$$

$$\mathbf{V}_1 = Z\,\mathbf{I}_1 = 22.36\angle -63.43^\circ \times 13\angle 20^\circ = \mathbf{290.7\angle -43.43^\circ\ V}$$

Check by KVL: $\mathbf{V}_R + \mathbf{V}_L + \mathbf{V}_C = 211.1 - j199.9 = 290.7\angle -43.43^\circ$ V.

(iii)

$$\mathbf{I}_1^* = \mathbf{13\angle -20^\circ\ A}$$

$$S = \mathbf{V}_1\mathbf{I}_1^* = 290.7\angle -43.43^\circ \times 13\angle -20^\circ = 3779\angle -63.43^\circ = \mathbf{1690 - j3380\ VA}$$

$$P = \mathbf{1690\ W}, \qquad Q = \mathbf{-3380\ VAr}\ \text{(the load supplies reactive power: net capacitive)}$$

Check: $P = I^2R = 13^2 \times 10 = 1690$ W and
$Q = I^2(X_L - X_C) = 169 \times (-20) = -3380$ VAr. The power factor is
$\cos 63.43^\circ = 0.4472$ leading.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [9 marks]

(i) Use $\mathcal{R} = l/(\mu_0\mu_r A)$ with $A = 20 \times 10^{-4}$ m² and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

$$\mathcal{R}_G = \frac{g}{\mu_0 A} = \frac{1.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 20 \times 10^{-4}} = \mathbf{596.8\ kA/Wb}$$

$$\mathcal{R}_C = \frac{l_c}{\mu_0\mu_r A} = \frac{0.5}{4\pi \times 10^{-7} \times 3000 \times 20 \times 10^{-4}} = \mathbf{66.31\ kA/Wb}$$

The two are in series:

$$L = \frac{N^2}{\mathcal{R}_G + \mathcal{R}_C} = \frac{32^2}{663.1 \times 10^3} = \mathbf{1.544\ mH}$$

The gap is 1.5 mm out of 500 mm of path, but it has 9 times the reluctance of the core. (The course uses the full 50 cm for the core. Subtracting the gap changes $\mathcal{R}_C$ by 0.3% and does not change the answer at 4 figures.)

(ii) The energy density in the gap is $B^2/(2\mu_0)$ J/m³, and the gap volume is Ag.

$$W_g = \frac{\hat{B}^2}{2\mu_0} A\,g = \frac{1.8^2}{2 \times 4\pi \times 10^{-7}} \times 20 \times 10^{-4} \times 1.5 \times 10^{-3} = \mathbf{3.867\ J}$$

(The total energy $\frac{1}{2}LI^2$ at 1.8 T is 4.297 J. The question asks for the **gap only**, so 3.867 J.)

(iii) Split first: $P_h = 0.3 \times 12 = 3.6$ W and $P_e = 0.7 \times 12 = 8.4$ W. Then scale each with its own law, $P_h \propto \hat{B}^2 f$ and $P_e \propto \hat{B}^2 f^2$.

$$P_h' = 3.6 \times \left(\frac{1.2}{1.5}\right)^2 \times \frac{100}{60} = \mathbf{3.84\ W}$$

$$P_e' = 8.4 \times \left(\frac{1.2}{1.5}\right)^2 \times \left(\frac{100}{60}\right)^2 = \mathbf{14.93\ W}$$

$$P_{FE}' = 3.84 + 14.93 = \mathbf{18.77\ W}$$

2b) Transformer Parameters [5 marks]

All values are referred to the **primary**, because both tests were measured on the primary side.

SC test (series branch; the shunt branch is negligible at 27 V):

$$|Z| = \frac{V_{SC}}{I_{SC}} = \frac{27}{9.7} = 2.784\ \Omega$$

$$R = R_1 + R_2' = \frac{P_{SC}}{I_{SC}^2} = \frac{55}{9.7^2} = \mathbf{0.5845\ \Omega}$$

$$X = X_1 + X_2' = \sqrt{|Z|^2 - R^2} = \sqrt{2.784^2 - 0.5845^2} = \mathbf{2.721\ \Omega}$$

OC test (shunt branch at rated voltage):

$$R_C = \frac{V_{OC}^2}{P_{OC}} = \frac{240^2}{45} = \mathbf{1280\ \Omega}$$

$$Q_{OC} = \sqrt{(V_{OC}I_{OC})^2 - P_{OC}^2} = \sqrt{312^2 - 45^2} = 308.7\ \text{VAr}$$

$$X_M = \frac{V_{OC}^2}{Q_{OC}} = \frac{240^2}{308.7} = \mathbf{186.6\ \Omega}$$

Sanity check: rated primary current is $2400/240 = 10$ A, so the SC test was run near rated current, and the SC voltage is 11% of rated.

2c) Transformer Analysis [6 marks]

(i) KVL from the primary terminals, through the series branch, to the load. The shunt branch sits directly across $\mathbf{V}_1$, so it does not appear.

$$\boxed{\mathbf{V}_2' = \mathbf{V}_1 - \mathbf{I}_2' \left[(R_1 + R_2') + j(X_1 + X_2')\right]}$$

Use $\mathbf{I}_2'$, **not** $\mathbf{I}_1$. $\mathbf{I}_1$ also includes the current in R_C and X_M.

(ii) With R_L' connected, $\mathbf{V}_1$ is across the series branch plus the load, so

$$\boxed{\mathbf{V}_2' = \mathbf{V}_1 \frac{R_L'}{R_L' + R_1 + R_2' + j(X_1 + X_2')}}$$

(iii) Lagging.

- The series path has reactance $X_1 + X_2'$, so the load current $\mathbf{I}_2' = \mathbf{V}_1/(R_L' + R + jX)$ **lags** $\mathbf{V}_1$.
- The load is purely resistive, so $\mathbf{V}_2' = R_L'\mathbf{I}_2'$ is **in phase with** $\mathbf{I}_2'$.
- Therefore $\mathbf{V}_2'$ lags $\mathbf{V}_1$ by the angle $\tan^{-1}\left[X/(R_L' + R)\right]$.

(The official sheet writes $\mathbf{V}_1'$ for the primary voltage. On the primary side $\mathbf{V}_1' = \mathbf{V}_1$.)

Question 3. AC and Induction Machines [20 marks]

3a) AC Machines [6 marks]

(i) The rated speed must be just below a synchronous speed $n_S = 60f/p$ (with p = pole-pairs).

$$p = \frac{60f}{n} = \frac{3000}{560} = 5.36 \Rightarrow p = 5, \qquad n_S = \frac{60 \times 50}{5} = 600 \text{ rpm}$$

So $p = 5$ pole-pairs, **10 poles**.

(ii) Induction machine, because the rated speed (560 rpm) is not a synchronous speed. It is below $n_S = 600$ rpm, so it runs with slip. A synchronous machine runs at exactly n_S.

(iii) The rotor copper loss is sP_{AG}, so even with no other losses $P_{OUT} \leq (1 - s)P_{AG}$.

$$s = \frac{n_S - n}{n_S} = \frac{600 - 560}{600} = 0.06667, \qquad \eta_{max} = 1 - s = \frac{n}{n_S} = \frac{560}{600} = \mathbf{93.33\%}$$

(iv) On the linear part of the curve, $\tau \propto (n_S - n)$, the slip speed. Rated slip speed is $600 - 560 = 40$ rpm, so at one-quarter torque it is 10 rpm.

$$n = 600 - \frac{40}{4} = \mathbf{590\ rpm}$$

$$\frac{P}{P_{rated}} = \frac{\tau\,\omega}{\tau_{rated}\,\omega_{rated}} = 0.25 \times \frac{590}{560} = \mathbf{26.34\%}$$

3b) Induction Machine Torque vs Speed [5 marks]

(i) From the simplified circuit, maximum torque occurs where R_2'/s equals the magnitude of the rest of the series impedance:

$$s_{max} = \frac{R_2'}{\sqrt{R_1^2 + (X_1 + X_2')^2}}$$

For maximum torque at standstill, $s_{max} = 1$, so the total rotor resistance must equal that magnitude:

$$R_2' + R_{EXT}' = \sqrt{0.4^2 + (1.3 + 1.5)^2} = 2.828\ \Omega$$

$$R_{EXT}' = 2.828 - 0.5 = \mathbf{2.328\ \Omega}\ (\text{referred to the stator})$$

X_M and R_C are not needed: in the simplified circuit they sit across the supply and do not affect I_2'.

(ii) Key facts for the sketch:

- All three curves go to **zero torque at** n_S.
- They have the **same peak torque**. τ_{max} does not depend on R_2'.
- The peak moves to lower speed as resistance increases: $s_{max} = 0.177$ (peak at $0.82\,n_S$) for $R_{EXT} = 0$; $s_{max} = 0.588$ ($0.41\,n_S$) for 0.5 pu (1.164 Ω); $s_{max} = 1$ (peak **at** $n = 0$) for 1 pu.
- Starting torque rises with R_{EXT}. Relative to the 1 pu curve it is 37% (0 pu), 89% (0.5 pu) and 100% (1 pu).

Torque against speed for the Jun23 Q3b machine with three values of external rotor resistance (simplified circuit, torque in relative units).

3c) Induction Machine Torque Calculation [9 marks]

(i) $p = 60f/n = 3000/1450 = 2.07 \Rightarrow p = 2$ pole-pairs, **4 poles**.

$$n_S = \frac{60f}{p} = \frac{3000}{2} = \mathbf{1500\ rpm}$$

(ii) Generator, because $n = 1520 > n_S = 1500$ rpm, so $s = (1500 - 1520)/1500 = -0.01333 < 0$. The rotor is driven faster than the field, and R_2'/s is negative.

(iii) Per-phase quantities (star): $V_1 = 415/\sqrt{3} = 239.6$ V.

$$s = \frac{1500 - 1450}{1500} = 0.03333, \qquad \frac{R_2'}{s} = \frac{0.2}{0.03333} = 6.0\ \Omega, \qquad \omega_S = \frac{2\pi \times 1500}{60} = 157.08\ \text{rad/s}$$

Torque formula:

$$\tau = \frac{3V_1^2\,(R_2'/s)}{\omega_S\left[(R_1 + R_2'/s)^2 + (X_1 + X_2')^2\right]} = \frac{3 \times 239.6^2 \times 6.0}{157.08 \times \left[6.3^2 + 5.0^2\right]} = \mathbf{101.7\ Nm}$$

(iv) Ignoring the shunt branch, $\mathbf{I}_1 = \mathbf{I}_2'$:

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{(R_1 + R_2'/s) + j(X_1 + X_2')} = \frac{239.6\angle 0^\circ}{6.3 + j5.0} = \frac{239.6\angle 0^\circ}{8.043\angle 38.44^\circ} = \mathbf{29.79\angle -38.44^\circ\ A}$$

Check: $P_{AG} = 3I_2'^2 R_2'/s = 3 \times 29.79^2 \times 6.0 = 15.97$ kW and $\tau = P_{AG}/\omega_S = 101.7$ Nm, matching (iii).

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

(i) Use the OC and SC readings at the **same field current**, with the OC voltage converted to a phase value.

$$|Z_S| = \frac{V_{OC}/\sqrt{3}}{I_{SC}} = \frac{4600/\sqrt{3}}{720} = 3.689\ \Omega$$

$$X_S = \sqrt{|Z_S|^2 - R^2} = \sqrt{3.689^2 - 0.3^2} = \mathbf{3.676\ \Omega}$$

(For reference, $Z_B = 14^2/150 = 1.307\ \Omega$, so $X_S = 2.81$ pu. The question asks for ohms.)

(ii) Bases: $S_B = 150$ MVA, $V_B = 14$ kV. At rated voltage, take the terminal voltage as reference:

$$\mathbf{V} = \mathbf{1.0\angle 0^\circ\ pu}$$

$$S = \frac{100 + j50}{150} = 0.6667 + j0.3333\ \text{pu}$$

In per-unit, $S = \mathbf{V}\mathbf{I}^*$, so

$$\mathbf{I} = \left(\frac{S}{\mathbf{V}}\right)^* = 0.6667 - j0.3333 = \mathbf{0.7454\angle -26.57^\circ\ pu}$$

The current lags because the generator is delivering reactive power (pf 0.894 lag). In amps this is $0.7454 \times 6186 = 4611$ A.

4b) Power Factor Correction [7 marks]

(i) 200 kW is the **output**. Input:

$$P_{im} = \frac{P_{out}}{\eta} = \frac{200}{0.85} = \mathbf{235.3\ kW}$$

$$Q_{im} = P_{im}\tan(\cos^{-1}0.83) = 235.3 \times 0.6720 = \mathbf{158.1\ kVAr}\ \text{(absorbed)}$$

(ii)

$$P_{syn} = \frac{200}{0.90} = \mathbf{222.2\ kW}$$

The 250 kVA rating limits $|S|$. The rest of the rating is available as reactive power, which an over-excited synchronous motor can **supply** (leading pf):

$$Q_{syn,max} = \sqrt{S_{rated}^2 - P_{syn}^2} = \sqrt{250^2 - 222.2^2} = \mathbf{114.5\ kVAr}\ \text{(supplied)}$$

(iii) Add the plant's powers with signs (absorbed reactive power positive):

$$P = 235.3 + 222.2 = 457.5\ \text{kW}, \qquad Q = 158.1 - 114.5 = 43.6\ \text{kVAr}$$

$$\text{pf} = \frac{P}{\sqrt{P^2+Q^2}} = \frac{457.5}{\sqrt{457.5^2 + 43.6^2}} = \mathbf{0.9955\ lag}$$

It is still lagging, because the synchronous motor cannot cancel all 158.1 kVAr. For comparison, the original plant (two induction motors) ran at 0.83 lag.

4c) Short-Answer Questions [8 marks]

(i) Power-factor correction. Many loads, such as induction motors and transformers, draw reactive power as well as real power. Reactive power is energy that oscillates back and forth between the supply and the load each cycle. It does no useful work, but it increases the current. This extra current loads up generators, transformers and lines, uses up their rating, and causes extra copper losses and voltage drop. Power-factor correction means adding a source of reactive power close to the load, usually a capacitor bank connected in parallel with it (or an over-excited synchronous machine). The capacitor generates the reactive power the load needs, so less reactive power is drawn from the supply. The supply current

falls for the same real power, the power factor moves towards one, and losses and charges for poor power factor are reduced.

(ii) Transformers and induction machines. *Similarities:* both can operate from a three-phase AC supply. Both have a primary (stator) winding and a secondary (rotor) winding coupled by a magnetic core. Both have nearly the same equivalent circuit, with winding resistances, leakage reactances, a magnetising reactance and a core-loss resistance, and both can be analysed per phase with a simplified circuit. *Differences:* in an induction machine the secondary winding is on the rotor, which turns, and it is short-circuited rather than feeding a load. The rotor frequency is the slip times the stator frequency, so it differs from the supply frequency. An induction machine also has an airgap, so its magnetising current is much larger. In the induction machine equivalent circuit, the referred rotor resistance is divided by the slip. This term represents both the rotor copper loss and the mechanical output power.

(iii) Perfectly conducting rotor. The rotor resistance would be zero. There would be no rotor copper loss, so efficiency would be higher. The rotor would carry whatever current is needed at almost exactly synchronous speed. The torque–speed curve would become vertical at synchronous speed, like a synchronous machine, so the speed would hardly change with load. But the machine would produce no torque away from synchronous speed, so it would have zero starting torque and could not start on its own. (This is the limit of the Q3b sketch as rotor resistance goes to zero: the peak moves right up to synchronous speed.)